



EXAM PAPERS PRACTICE

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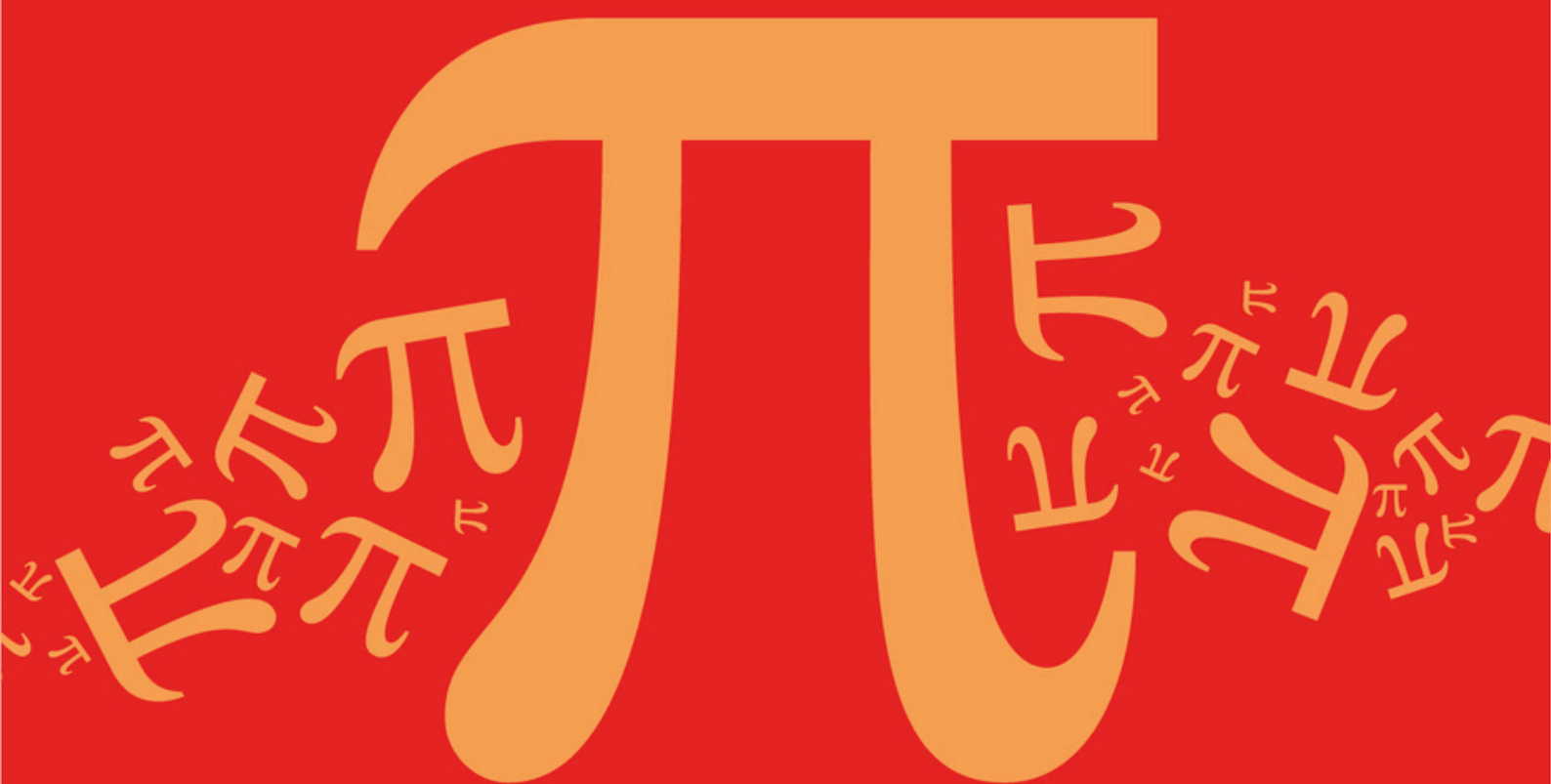
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AQA A Level Further Mathematics (7367)



Topic

B: Complex Numbers

Sub topic

B11: Geometric Problems

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

B: Complex Numbers

Sub topic

B11: Geometric Problems

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

- (a) Two points, A and B , on an Argand diagram are represented by the complex numbers $2 + 3i$ and $-4 - 5i$ respectively. Given that the points A and B are at the ends of a diameter of a circle C_1 , express the equation of C_1 in the form $|z - z_0| = k$.

(4)

- (b) A second circle, C_2 , is represented on the Argand diagram by the equation $|z - 5 + 4i| = 4$. Sketch on one Argand diagram both C_1 and C_2 .

(3)

- (c) The points representing the complex numbers z_1 and z_2 lie on C_1 and C_2 respectively and are such that $|z_1 - z_2|$ has its maximum value. Find this maximum value, giving your answer in the form $a + b\sqrt{5}$.

(5)

(Total 12 marks)

Q2.

- (a) A circle C in the Argand diagram has equation

$$|z + 5 - i| = \sqrt{2}$$

Write down its radius and the complex number representing its centre.

(2)

- (b) A half-line L in the Argand diagram has equation

$$\arg(z + 2i) = \frac{3\pi}{4}$$

Show that $z_1 = -4 + 2i$ lies on L .

(2)

- (c) (i) Show that $z_1 = -4 + 2i$ also lies on C .

(1)

(ii) Hence show that L touches C .

(3)

(iii) Sketch L and C on one Argand diagram.

(2)

- (d) The complex number z_2 lies on C and is such that $\arg(z_2 + 2i)$ has as great a value as possible.

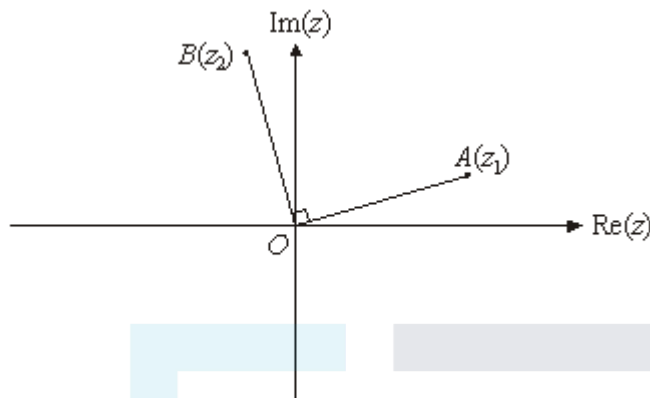
Indicate the position of z_2 on your sketch.

(2)

(Total 12 marks)

Q3.

The sketch shows an Argand diagram. The points A and B represent the complex numbers z_1 and z_2 respectively. The angle $AOB = 90^\circ$ and $OA = OB$.



- (a) Explain why $z_2 = iz_1$. (2)
- (b) On a **single** copy of the diagram, draw:
- (i) the locus L_1 of points satisfying $|z - z_2| = |z - z_1|$ (2)
- (ii) the locus L_2 of points satisfying $\arg(z - z_2) = \arg z_1$. (3)
- (c) Find, in terms of z_1 , the complex number representing the point of intersection of L_1 and L_2 . (2)

(Total 9 marks)

Q4.

The complex numbers z_1 and z_2 are given by

$$z_1 = \frac{1+i}{1-i} \quad \text{and} \quad z_2 = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

- (a) Show that $z_1 = i$. (2)
- (b) Show that $|z_1| = |z_2|$. (2)
- (c) Express both z_1 and z_2 in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. (3)
- (d) Draw an Argand diagram to show the points representing z_1 , z_2 and $z_1 + z_2$. (2)

- (e) Use your Argand diagram to show that

$$\tan \frac{5}{12} \pi = 2 + \sqrt{3}$$

(3)

(Total 12 marks)

Q5.

The complex number z satisfies the relation

$$|z + 4 - 4i| = 4$$

- (a) Sketch, on an Argand diagram, the locus of z .

(3)

- (b) Show that the greatest value of $|z|$ is $4(\sqrt{2} + 1)$.

(3)

- (c) Find the value of z for which

$$\arg(z + 4 - 4i) = \frac{1}{6} \pi$$

Give your answer in the form $a + ib$.

(3)

(Total 9 marks)

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